

YEAR 9

- 1) Ella wants a rectangle with
- A perimeter of 100cm and
 - The largest possible area

What are the dimensions of the rectangle that satisfies her condition?

Identify the following equation as being linear or non-linear. (Question 3-4)

- 2) $3x + 2y = 7$
- 3) $x^2 - 3x + 4 = y$
- 4) $9x - 2 = \frac{y}{2}$
- 5) Find the length of the diagonal of a rectangle, D given that the length of the rectangle is 10.7cm and the diagonal makes an angle of 39° with the longer side. Correct your answer to 1 decimal place
- 6) Adams earns £5, 000 per week. Each week, he spends £3x on food, £2x on rent and saves the rest. How much does he save in a week?

Demi charges £12 to cut and rake one lawn. She earned £1176 for the summer. (Question 7-8)

- 7) Model this situation using an equation.
- 8) Solve the equation to determine how many lawns she cut and raked over the summer
- 9) $\angle 1$, $\angle 2$, and $\angle 3$ are adjacent angles, with $\angle 1$ supplementary to $\angle 2$ and $\angle 2$ complementary to $\angle 3$. If $m\angle 1 = (8x + 3)^\circ$ and $m\angle 2 = (5x - 18)^\circ$, find $m\angle 3$

Let $A = \{1, 2, 3, 8, 10, 12\}$ and $B = \{6, 5, 4, 3, 2, 10\}$ write down the set.

- 10) $A \cap B$
- 11) $A \cup B$
- 12) Solve the inequality $3x - 7 > 11$. Write your answer using set notation
- 13) If $(A + B)^2 = A^2 + 2AB + B^2$ then what can we say about A and B?
(Assume AB and BA exists)

- 14) If the sides of a right triangle are in the ratio 3:4:5, and the shortest side is 9cm, find the length of the hypotenuse.
- 15) If A is a $n \times m$ matrix such that AB and BA are both defined, then order of B is?
- 16) A shopkeeper purchased 200 bulbs for £10 each. However 5 bulbs were fused and had to be thrown away. The remaining were sold at £12 each. Find the gain or loss%.
- 17) Find the sine of an angle a in a right triangle, where the opposite sides is 6 units and the hypotenuse is 10 units.
- 18) The price of a motor bike was £60, 000 last year. It has increased by 15% this year. What is the price now?

$$\xi = \{103, 104, 105, 109, 110, 112, 114\}$$

A = even number from ξ

$$B = \{103, 112, 114\}$$

State which numbers belong to the following:

19) $A \cup B$

20) A'